

ADHITHYA R

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CAREER OBJECTIVE

Electronics and Communication Engineering student (CGPA 8.4) with demonstrated experience in embedded systems design, PCB layout, and IoT-based automation. Proficient in STM32 and Arduino microcontroller ecosystems, KiCad schematic capture, and C/C++ firmware development. Seeking a VLSI / Embedded Systems internship to contribute to hardware design, RTL verification, and SoC-level integration while deepening expertise in digital design flows.

EDUCATION

B.E. – Electronics and Communication Engineering

Sep 2024 – May 2028

K.S.R College of Engineering, Tiruchengode | CGPA: 8.35 / 10.0

Higher Secondary Certificate (HSC)

2023 – 2024

Vijayalakshmi Matriculation Higher Secondary School, Coimbatore | 81.34%

Secondary School Leaving Certificate (SSLC)

2021 – 2022

Vijayalakshmi Matriculation Higher Secondary School, Coimbatore | 91.3%

TECHNICAL SKILLS

Microcontrollers & SoCs: STM32 (Cortex-M) • Arduino Uno/Nano • Esp-32 • Raspberry Pi 4

Hardware Design: KiCad (Schematic & PCB Layout) • MultiSim (Simulation) • Vivado (FPGA / RTL)

Programming: C / C++ • Embedded C (HAL / LL drivers) • Basic Python • Verilog HDL

Protocols & Interfaces: UART • SPI • I2C • Bluetooth (HC-05) • GPIO

Domain Skills: Sensor Interfacing • PCB Design & Fabrication • IoT Systems • Image Processing

Tools & Platforms: Git / GitHub • VS Code • Arduino IDE • STM32CubeIDE • Keil µVision • Xilinx

PROJECTS

STM32 Custom Microcontroller PCB Design

2025

Designed a production-ready STM32-based microcontroller board in KiCad featuring 16 GPIO pins, USB Type-C power input, SWD debug header, and decoupling networks.

- Performed schematic capture, component placement, and 2-layer routing following IPC-2221 guidelines.
- Exported Gerber files and BOM; validated design rule checks (DRC) with zero critical errors.

GitHub: github.com/adhithya641/STM32_Embedded_Controller

Autonomous Pesticide Spraying Robot

2025

Designed and built an autonomous agricultural robot for targeted pesticide application using computer vision-based weed detection.

- Integrated AI-Thinker ESP32-CAM for real-time image capture and weed classification using edge inference.
- Used HC-05 Bluetooth module for wireless command interface; controlled pump actuators via Arduino Uno GPIO.
- Reduced chemical usage by targeting only detected weed regions, demonstrating precision agriculture concepts.

Automated Attendance Monitoring System

2025

Built a vision-based attendance system using Raspberry Pi 4 and Logitech webcam with OpenCV face recognition.

- Captured and logged student images per period; automated attendance marking with timestamp-based records stored in CSV.
- Eliminated manual roll-call, demonstrating embedded Linux application development on Raspberry Pi.

INTERNSHIP

VLSI Based Intern – Clarozon Technologies

July 2025

- Completed a 2-week intensive internship covering Verilog, Digital, RTL, and FPGA.
- Gained exposure to work on FPGA based projects.

HACKATHONS & COMPETITIONS

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- Smart India Hackathon 2025 (SIH-2025) — College representative on the Rural Education theme.
 - LUNARA'26 — National-level technical event at Bannari Amman Institute of Technology.
 - CODEATHON — Coding competition hosted at K.S.R College of Engineering.
 - Conducted workshop on Computer Networks at K.S.R College of Engineering

CERTIFICATIONS

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- NPTEL — Artificial Intelligence (IIT Madras)
 - NPTEL — Internet of Things (IIT Madras)
 - Eduskills — Microchip Embedded Systems Certification
 - NXT Wave — Full-Stack Development

LANGUAGES

English (Fluent) • Tamil (Fluent)